

Safety Data Sheet

According to Notification of Thailand Ministry of Industry B.E. 2555 (2012)



Methanol

Version 1.0

Release Date: 2022/07/21

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : Methanol
Synonyms : Methyl hydrate; Wood spirit; Methyl hydroxide
CAS-No. : 67-56-1

Recommended use of the chemical and restrictions on use

Recommended use : For industrial use only. Feedstock, fuel, solvent and industrial solvent for commercial products.
Restrictions on use : Do not use this product other than stated in recommended use without first seeking the advice of the principal supplier.

Manufacturer or supplier's details

Headquarters

Company : PETRONAS Chemicals Group Berhad
Address : Tower 1, PETRONAS Twin Towers,
Kuala Lumpur City Centre,
50088 Kuala Lumpur
Malaysia

Plant Site

Company : PETRONAS Chemicals Methanol Sdn Bhd
Address : Rantau Rantau Industrial Estate,
87000 Federal Territory Labuan
Labuan, Malaysia

Emergency telephone number : 24-Hour Emergency Contact: 001-800-13-203-9987

2. HAZARDS IDENTIFICATION

GHS Classification

Flammable liquids : Category 2
Acute toxicity (Oral) : Category 3
Acute toxicity (Inhalation) : Category 3
Acute toxicity (Dermal) : Category 3
Specific target organ toxicity - single exposure : Category 1

GHS label elements

Hazard pictograms



Signal word : Danger
Hazard statements : H225 Highly flammable liquid and vapour.
H301 + H311 + H331 Toxic if swallowed, in contact with skin or if inhaled.
H370 Causes damage to organs.
Precautionary statements : **Prevention:**
P210 Keep away from heat/sparks/open flames/hot surfaces.
No smoking.

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P233 Keep container tightly closed.
P240 Ground/bond container and receiving equipment.
P241 Use explosion-proof electrical/ ventilating/ lighting equipment.
P242 Use only non-sparking tools.
P243 Take precautionary measures against static discharge.
P260 Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
P264 Wash skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves/ eye protection/ face protection.

Response:

P301 + P310 + P330 IF SWALLOWED: Immediately call a POISON CENTER/doctor. Rinse mouth.

P303 + P361 + P353 IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower.

P304 + P340 + P311 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/ physician.

P307 + P311 IF exposed: Call a POISON CENTER or doctor/ physician.

P363 Wash contaminated clothing before reuse.

P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction.

Storage:

P403 + P233 + P235 Store in a well-ventilated place. Keep container tightly closed. Keep cool.

P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

No information available.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Substance
Chemical nature : Methanol
Substance name : Methyl hydrate; Wood spirit; Methyl hydroxide
CAS-No. : 67-56-1

Components

Chemical name	CAS-No.	Concentration (%)
Methanol	67-56-1	>= 90 -<= 100

4. FIRST AID MEASURES

If inhaled : If unconscious, place in recovery position and seek medical

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	advice.
In case of skin contact	: If symptoms persist, call a medical doctor. If on skin, rinse well with water.
In case of eye contact	: If on clothes, remove clothes. Flush eyes with water as a precaution. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	: Keep respiratory tract clear. Do NOT give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. Take victim immediately to hospital. If symptoms persist, call a medical doctor.
General advice	: Induce vomiting immediately and call a medical doctor. Move out of dangerous area. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended. Consult a medical doctor.
Most important symptoms and effects, both acute and delayed	: Central nervous system effects of headache, lethargy, confusion, blurred vision, photophobia and blindness.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	: Carbon dioxide (CO ₂) Alcohol-resistant foam Dry chemical
Unsuitable extinguishing media	: High volume water jet
Specific hazards during firefighting	: Do not allow run-off from fire fighting to enter drains or water courses.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed off in accordance with local regulations. For safety reasons in case of fire, cans should be stored separately in closed containments. Use a water spray to cool fully closed containers.
Special protective equipment for firefighters	: Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Use personal protective equipment. Remove all sources of ignition. Ensure adequate ventilation. Evacuate personnel to safe areas. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.
Environmental precautions	: If the product contaminates rivers and lakes or drains inform

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- Methods and materials for containment and cleaning up : respective authorities.
Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
: Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapours).
Use only explosion-proof equipment.
Do not spray on a naked flame or any incandescent material.
Keep away from open flames, hot surfaces and sources of ignition.
- Advice on safe handling : Normal measures for preventive fire protection.
: Avoid contact with skin and eyes.
Container may be opened only under exhaust ventilation hood.
Avoid formation of aerosol.
Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Open drum carefully as content may be under pressure.
Dispose of rinse water in accordance with local and national regulations.
- Conditions for safe storage : Take precautionary measures against static discharges.
: Keep container tightly closed in a dry and well-ventilated place.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.
No smoking.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
- Further information on storage stability : No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION**Components with workplace control parameters**

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis

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Methanol	67-56-1	TWA	200 ppm 260 mg/m ³	TH OEL
		TWA	200 ppm	ACGIH
		STEL	250 ppm	ACGIH

Biological occupational exposure limits

Components	CAS-No.	Control parameters	Biological specimen	Sampling time	Permissible concentration	Basis
Methanol	67-56-1	Methanol	Urine	End of shift (As soon as possible after exposure ceases)	15 mg/l	ACGIH BEI

Personal protective equipment

Respiratory protection : In the case of vapour formation use a respirator with an approved filter.

Hand protection

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water.
Tightly fitting safety goggles

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Hygiene measures : Avoid contact with skin, eyes and clothing.
Wash hands before breaks and immediately after handling the product.
When using do not eat or drink.
When using do not smoke.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Liquid
Colour	: Colourless
Odour	: Alcohol-like
Odour Threshold	: 4.26 ppm
pH	: No data available
Melting point/freezing point	: -98 °C
Boiling point/boiling range	: 65 °C
Flash point	: 9 °C Method: ASTM D 56, Tag closed cup
Evaporation rate	: No data available
Upper explosion limit / Upper flammability limit	: 50 %(V)665.000 mg/m ³
Lower explosion limit / Lower flammability limit	: 6 %(V)80.000 mg/m ³
Vapour pressure	: 126 hPa (20 °C)
Relative vapour density	: 1.10
Relative density	: 1.01 (20 °C)
Density	: 0.79 g/cm ³ (20 °C)

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Solubility(ies)	
Water solubility	: Completely miscible
Partition coefficient: n-octanol/water	: No data available
Auto-ignition temperature	: No data available
Decomposition temperature	: No data available
Viscosity	
Viscosity, dynamic	: No data available
Viscosity, kinematic	: 0.3 mm ² /s (100 °C)
Molecular weight	: 32.04 g/mol

10. STABILITY AND REACTIVITY

Reactivity	: Hazardous polymerisation does not occur.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, sparks, flame and build-up of static electricity.
Incompatible materials	: Strong oxidizing agents
Hazardous decomposition products	: Fumes, smoke, carbon monoxide

11. TOXICOLOGICAL INFORMATION**Acute toxicity****Components:****Methanol:**

Acute oral toxicity	: Remarks: No data available
Acute inhalation toxicity	: Remarks: No data available
Acute dermal toxicity	: Remarks: No data available

Skin corrosion/irritation**Components:****Methanol:**

Remarks	: No data available
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Serious eye damage/eye irritation**Components:****Methanol:**

Remarks	: No data available
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Respiratory or skin sensitisation**Components:****Methanol:**

Exposure routes	: Inhalation
Remarks	: No data available

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Exposure routes : Skin contact
Remarks : No data available

Germ cell mutagenicity

Components:

Methanol:

Germ cell mutagenicity - : No data available
Assessment

Carcinogenicity

Components:

Methanol:

Carcinogenicity - : No data available
Assessment

Reproductive toxicity

Components:

Methanol:

Reproductive toxicity - : Animal studies suggest toxicity to reproduction.
Assessment

STOT - single exposure

Components:

Methanol:

Target Organs : Central Nervous System, Eyes, Lungs
Assessment : Affects central nervous system, toxicity to optic nerves which may lead to blindness and respiratory irritation., 4-10ml may cause permanent blindness, lethal dose : 30-200ml

STOT - repeated exposure

Components:

Methanol:

Remarks : No data available

Aspiration toxicity

Components:

Methanol:

Statement on Aspiration Tox. : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

Methanol:

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 15,400 mg/l
Exposure time: 96 h

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Toxicity to daphnia and other aquatic invertebrates	Test Type: flow-through test
	: EC50 (Daphnia magna (Water flea)): 18,260 mg/l
	Exposure time: 96 h
Toxicity to algae	Test Type: semi-static test
	: EC50 (Pseudokirchneriella subcapitata (green algae)): 22,000 mg/l
	Exposure time: 96 h
Toxicity to fish (Chronic toxicity)	Test Type: static test
	: NOEC (Pimephales promelas (fathead minnow)): 446.7 mg/l
	Exposure time: 28 d
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	: NOEC (Daphnia magna (Water flea)): 208 mg/l
	Exposure time: 21 d
Toxicity to microorganisms	: IC50 (activated sludge): > 1,000 mg/l
	Exposure time: 3 h
	Test Type: static test
	Remarks: Oxygen consumption, sealed serum bottles. Mimic the standard ATA. It uses cell concentrations typical of the AFNOR and ETAD standard assays. Effects of cell growth were measured similar to standard BOD test.

Persistence and degradability**Components:****Methanol:**

Biodegradability : Result: Readily biodegradable.

Bioaccumulative potential**Components:****Methanol:**

Bioaccumulation : Bioconcentration factor (BCF): < 10

Partition coefficient: n-octanol/water : log Pow: -0.77

Mobility in soil**Components:****Methanol:**Mobility : Medium: Soil
Remarks: Very high mobility.**Other adverse effects****Components:****Methanol:**

Environmental fate and pathways : Exist solely in vapor phase and miscible with water.

13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues	: Do not dispose of waste into sewer.
	Do not contaminate ponds, waterways or ditches with chemical or used container.
	Send to a licensed waste management company.
Contaminated packaging	: Empty remaining contents.
	Dispose of as unused product.
	Do not re-use empty containers.

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Do not burn, or use a cutting torch on, the empty drum.

14. TRANSPORT INFORMATION**International Regulations****UNRTDG**

UN number : UN 1230
Proper shipping name : METHANOL
Class : 3
Subsidiary risk : 6.1
Packing group : II
Labels : 3 (6.1)

IATA-DGR

UN/ID No. : UN 1230
Proper shipping name : Methanol
Class : 3
Subsidiary risk : 6.1
Packing group : II
Labels : Class 3 - Flammable liquids, Division 6.1 - Toxic substances
Packing instruction (cargo aircraft) : 364
Packing instruction (passenger aircraft) : 352

IMDG-Code

UN number : UN 1230
Proper shipping name : METHANOL
Class : 3
Subsidiary risk : 6.1
Packing group : II
Labels : 3 (6.1)
EmS Code : F-E, S-D
Marine pollutant : no

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Pollution category : Y
Ship type : 3

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

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The components of this product are reported in the following inventories:

CH INV	: On the inventory, or in compliance with the inventory.
TSCA	: On TSCA Inventory.
DSL	: All components of this product are on the Canadian DSL
AICS	: On the inventory, or in compliance with the inventory.
NZIoC	: On the inventory, or in compliance with the inventory.
ENCS	: On the inventory, or in compliance with the inventory.
ISHL	: On the inventory, or in compliance with the inventory.
KECI	: On the inventory, or in compliance with the inventory.
PICCS	: On the inventory, or in compliance with the inventory.
IECSC	: On the inventory, or in compliance with the inventory.

16. OTHER INFORMATION

Revision Date : 2022/07/21
 Sources of key data used to compile the Safety Data Sheet : ICOP CCHC

Full text of other abbreviations

(Q)SAR	-	(Quantitative) Structure Activity Relationship
ACGIH	-	American Conference of Governmental Industrial Hygienists
AICS	-	Australian Inventory of Chemical Substances
ANTT	-	National Agency for Transport by Land of Brazil
ASTM	-	American Society for the Testing of Materials
bw	-	Body weight
C	-	Ceiling
CCHC	-	Chemicals Classification and Hazard Communication
CEIL	-	Ceiling
ChV	-	Chronic Toxicity Value
CMR	-	Carcinogen, Mutagen or Reproductive Toxicant
CPR	-	Controlled Products Regulations
DIN	-	Standard of the German Institute for Standardisation
DSL	-	Domestic Substances List (Canada)
ECx	-	Concentration associated with x% response
ELx	-	Loading rate associated with x% response
EmS	-	Emergency Schedule
ENCS	-	Existing and New Chemical Substances (Japan)
ErCx	-	Concentration associated with x% growth rate response
ERG	-	Emergency Response Guide
GHS	-	Globally Harmonized System of Classification and Labelling of Chemicals
GLP	-	Good Laboratory Practice
IARC	-	International Agency for Research on Cancer
IATA	-	International Air Transport Association
IBC	-	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
IC50	-	Half maximal inhibitory concentration
ICAO	-	International Civil Aviation Organization
ICOP	-	Industry Code of Practice on Chemicals Classification and Hazard Communication
IECSC	-	Inventory of Existing Chemical Substances in China
IMDG	-	International Maritime Dangerous Goods

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IMO	-	International Maritime Organization
ISHL	-	Industrial Safety and Health Law (Japan)
ISO	-	International Organisation for Standardization
KECI	-	Korea Existing Chemicals Inventory
LC50	-	Lethal Concentration to 50 % of a test population
LD50	-	Lethal Dose to 50% of a test population (Median Lethal Dose)
MARPOL	-	International Convention for the Prevention of Pollution from Ships
MY PEL	-	Malaysian Permissible Exposure Limit
n.o.s.	-	Not Otherwise Specified
Nch	-	Chilean Norm
NITE	-	National Institute of Technology and Evaluation
NO(A)EC	-	No Observed (Adverse) Effect Concentration
NO(A)EL	-	No Observed (Adverse) Effect Level
NOELR	-	No Observable Effect Loading Rate
NOM	-	Official Mexican Norm
NTP	-	National Toxicology Program
NZIoC	-	New Zealand Inventory of Chemicals
OCSP	-	Office of Chemical Safety and Pollution Prevention
OECD	-	Organization for Economic Co-operation and Development
PBT	-	Persistent, Bioaccumulative and Toxic
PICCS	-	Philippines Inventory of Chemicals and Chemical Substances
REACH	-	Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
SADT	-	Self-Accelerating Decomposition Temperature
SDS	-	Safety Data Sheet
STEL	-	Short Term Exposure Limit
TCSI	-	Taiwan Chemical Substance Inventory
TDG	-	Transportation of Dangerous Goods
TSCA	-	Toxic Substances Control Act (United States)
TWA	-	Time Weighted Average
UN	-	United Nations
UNRTDG	-	United Nations Recommendations on the Transport of Dangerous Goods
UVCB	-	Unknown or Variable Composition, Complex Reaction Products and Biological Materials
vPvB	-	Very Persistent and Very Bioaccumulative
WHMIS	-	Workplace Hazardous Materials Information System
LLx	-	Loading rate associated with x% lethality effect

Disclaimer

To the best of PETRONAS knowledge, the information contained herein is accurate and reliable as of the date of publication. Any use of information provided should be undertaken in consultation with a professional with the appropriate expertise, as necessary in the circumstances. No warranty, express or implied, is given as to the quality, accuracy, reliability, applicability or completeness of the contents of this SDS. The information presented here pertains only to the product as shipped. It is the responsibility of the customer to ensure that any activities relating to the product comply with all federal, state or local laws. Any hazards associated with any product regulatory requirements are subject to change and may differ between various locations. Except to the extent required by law, re-publication or retransmission of this SDS, in whole or in part, is strictly prohibited. PETRONAS does not take responsibility for use, transportation, storage, handling, packaging or disposal of the product that is beyond our knowledge and recommendation by this SDS.

Product Stewardship Advisory:

PETRONAS aims to increase awareness of all the hazards associated with the storage,

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handling and use of its products. Thoroughly reviewing the accompanying Safety Data Sheets and disseminating the information to all dependent and interested parties is an essential part of any 'Responsible Care' programme.

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